

國立臺灣師範大學
理學院
地球科學系
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College of Science
National Taiwan Normal University
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原生恆星盤外圍絲狀結構之模型與觀測比較
Modeling Streamer Structures in Collapsing
Prestellar Cores

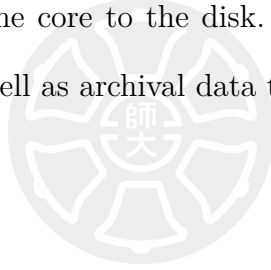
莊宗翰
Tsung-Han Chuang

指導教授：李悅寧 博士
Advisor: Yueh-Ning Lee, Ph.D.

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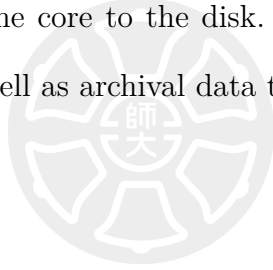
摘要

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.



Abstract

Streamers have been observed with high-resolution ALMA observations around many protoplanetary disks undergoing formation. They have been suggested to dominate the mass accretion budget from the collapsing prestellar core, and have important consequences on the disk dynamics. Understanding the formation of streamers is therefore important for explaining how protoplanetary disks receive mass from the envelope. We propose a model to explain the formation of streamer structures by considering density enhancements due to gravitational instability. We test our model against sources where streamers have been detected (ex., Per-emb-2 and Per-emb-50 observed with NOEMA) and fit for model parameters. This allows us to gain deeper insights into the physical origin of streamers and their role in mass transport from the core to the disk. This model can be applied to analyze new observations as well as archival data that show signs of streamers.



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Chapter 1

Introduction

1.1 The formation of cores

In a molecular cloud, a local density enhancement can trigger gravitational collapse, forming a prestellar core. As the collapse proceeds, the prestellar core becomes a protostellar core. During the early class 0 or class I stage, these systems are still deeply embedded within their envelope. Within this turbulent environment, small fluctuations can lead to gravitational instability, giving rise to an asymmetric, filamentary structure known as the streamer. Streamer is the density enhancement structures within the protostellar envelope. The key characteristics are that do not necessarily follow the Keplerian rotation. They have been considered as a channel to carry mass onto the disk.

1.2 The streamer in observations

Recently, several observations revealed asymmetric structures in the protostellar system (e.g., Pineda et al. 2020, Valdivia-Mena et al. 2022, Hsieh et al. 2023). The features of streamers are very diverse in morphology, do not always form in a disk plane, and have very weak density enhancement in observation.

1.3 Our work overview

The formation process of protoplanetary disks is still unrefined. Given their importance, we have developed a new model to characterize the streamer features and kinematics, focusing on gravitationally driven infalling and rotation.

1.3.1 Build up a model

1.3.2 Fitting with observations



Chapter 2

Methods

We build a model that can reproduce the observed streamer, by considering a self collapse density enhancement structure with a rotation velocity and evolve with time in 3D space.

2.1 Observations



To test our model against sources where streamers have been detected, we use five observational datasets: Per-emb-2, Per-emb-50, S CrA, HL Tau, and SVS 13A. For each source, we extract the position-position-velocity (PPV) data of the streamer from published NOEMA, ALMA observations. The fitting procedure described below is applied to each dataset to constrain the model parameters, including the density enhancement along the polar angle (θ), infall velocity (v_{inf}), rotation velocity (v_{rot}), accumulated rotation (ϕ) and inclination (i). The results are compared to the observed streamer morphology and kinematics to assess the validity of the model.

2.2 Formalism

The sonic length.

$$r_s = c_s T \quad (2.1)$$

Infall velocity

$$v_{\text{inf}} = -\sqrt{\frac{2GM_*}{r}} - 3.3c_s \quad (c_s = 200\text{m/s}) \quad (2.2)$$

The rotation velocity.

$$v_{\text{rot}} = r_s \omega \Omega_{\text{kep}} \left[\left(\frac{r}{r_s} \right)^{-1} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.3)$$

The accumulated rotation.

$$\phi = \phi_0 + t_s \omega \Omega_{\text{kep}} \sqrt{\frac{2c_s^3 t_s}{GM_*}} \left[\left(\frac{r}{r_s} \right)^{-\frac{1}{2}} + \left(\frac{r}{r_s} \right)^{-\frac{1}{3}} \right] \quad (2.4)$$

2.3 Coordinate transformation

Position

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases} \quad (2.5)$$

Infall velocity

$$\begin{cases} v_{x_{\text{inf}}} = v_{\text{inf}} \sin \theta \cos \phi \\ v_{y_{\text{inf}}} = v_{\text{inf}} \sin \theta \sin \phi \\ v_{z_{\text{inf}}} = v_{\text{inf}} \cos \theta \end{cases} \quad (2.6)$$

Rotation velocity

$$\begin{cases} v_x = -v_{\text{rot}} \sin \theta \sin \phi + v_{x_{\text{inf}}} \\ v_y = v_{\text{rot}} \sin \theta \cos \phi + v_{y_{\text{inf}}} \\ v_z = 0 + v_{z_{\text{inf}}} \end{cases} \quad (2.7)$$

2.4 Streamer on the sky in the PPV space

Position rotates by inclination. Inclination(*i*): backward (+), forward (-).

$$\begin{cases} x' = r \sin \theta \cos \phi \\ y' = y \cos i + z \sin i \\ z' = -y \sin i + z \cos i = r(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.8)$$

Velocity rotates by inclination

$$\begin{cases} v'_x = v_x \\ v'_y = v_y \cos i + v_z \sin i = v_{\text{rot}} \sin \theta \cos \phi \cos i + v_{\text{inf}}(\sin \theta \sin \phi \cos i + \cos \theta \sin i) \\ v'_z = -v_y \sin i + v_z \cos i = -v_{\text{rot}} \sin \theta \cos \phi \sin i + v_{\text{inf}}(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \end{cases} \quad (2.9)$$

θ corresponds to the initial polar angle of the streamer, ϕ is the initial azimuth angle, i is the inclination angle (zero inclination corresponds to an edge-on disk), ω is the ratio of Keplerian rotation velocity, and t_s is the collapsing time of the

particle within the streamer.

2.5 Fitting procedure

2.5.1 Global fitting of the full model

[(YN) Re-include the infall term. Perform full model fitting around the optimal parameters found in the first step.]

2.5.2 First fitting: regression to approximative model

[(YN) Describe how this is allowed by omitting the infall term.] Use X , Z , V_y to fit with observation data.

$$\begin{cases} X = r \sin \theta \cos \phi \\ Z = r(-\sin \theta \sin \phi \sin i + \cos \theta \cos i) \\ V_y = r \Omega_{\text{ref}} \left(\frac{r}{r_{\text{ref}}} \right)^{-\frac{4}{3}} \sin \theta \cos \phi \cos i + v_{\text{inf}}(\sin \theta \sin \phi \cos i + \cos \theta \sin i) \end{cases} \quad (2.10)$$

Fitting for the initial parameter.

$$\begin{aligned} \frac{V_y}{X} &= \frac{r \sin \theta \Omega_{\text{ref}} \left(\frac{r}{r_{\text{ref}}} \right)^{-\frac{4}{3}} \cos \phi \cos i}{r \sin \theta \cos \phi} = \Omega_{\text{ref}} \left(\frac{r}{r_{\text{ref}}} \right)^{-\frac{4}{3}} \cos i \\ \left(\frac{r}{r_{\text{ref}}} \right)^{-\frac{4}{3}} &= \frac{V_y}{X \Omega_{\text{ref}} \cos i} \\ \frac{X}{r} &= (\Omega_{\text{ref}} \cos i)^{-\frac{3}{4}} \frac{X}{r_{\text{ref}}} \left(\frac{V_y}{X} \right)^{\frac{3}{4}} \end{aligned} \quad (2.11)$$

$$\begin{aligned} X \cdot \left(\frac{V_y}{X} \right)^{\frac{3}{4}} &= r_{\text{ref}} \sin \theta \cos \phi (\Omega_{\text{ref}} \cos i)^{\frac{3}{4}} \\ Z \cdot \left(\frac{V_y}{X} \right)^{\frac{3}{4}} &= (\cos \theta \cos i - \sin \theta \sin \phi \sin i) r_{\text{ref}} (\Omega_{\text{ref}} \cos i)^{\frac{3}{4}} \end{aligned} \quad (2.12)$$

$$\begin{aligned}
A &= r_{\text{ref}}(\Omega_{\text{ref}} \cos i)^{\frac{3}{4}} \sin \theta \\
B &= -r_{\text{ref}}(\Omega_{\text{ref}} \cos i)^{\frac{3}{4}} \sin \theta \sin i \\
C &= r_{\text{ref}}(\Omega_{\text{ref}} \cos i)^{\frac{3}{4}} \cos \theta \cos i \\
k &= \frac{t}{\sin i} \\
\phi &= k(R_i + R_0)
\end{aligned} \tag{2.13}$$

$$\begin{aligned}
R_i &= \frac{V_y}{X} \\
X_i &= A \cos(k(R_i + R_0)) \\
Y_i &= B \sin(k(R_i + R_0)) + C
\end{aligned} \tag{2.14}$$

$$SEE^2 = \sum (X_i - A \cos(k(R_i + R_0))) + \sum (Y_i - B \sin(k(R_i + R_0)) - C) \tag{2.15}$$

$$\frac{1}{2} \frac{\partial SEE^2}{\partial A} = A \sum \cos^2(k(R_i + R_0)) - \sum X_i \cos(k(R_i + R_0))$$

$$\frac{1}{2} \frac{\partial SEE^2}{\partial B} = B \sum \sin^2(k(R_i + R_0)) - \sum Y_i \sin(k(R_i + R_0)) + C \sum \sin(k(R_i + R_0))$$

$$\frac{1}{2} \frac{\partial SEE^2}{\partial C} = \sum Y_i - \sum B \sin(k(R_i + R_0)) - NC$$

$$\begin{aligned} \frac{1}{2} \frac{\partial SEE^2}{\partial k} &= A \sum X_i R_i \sin(k(R_i + R_0)) - A^2 \sum R_i \cos(k(R_i + R_0)) \sin(k(R_i + R_0)) \\ &\quad - B \sum Y_i R_i \cos(k(R_i + R_0)) + B^2 \sum R_i \sin(k(R_i + R_0)) \cos(k(R_i + R_0)) \\ &\quad + BC \sum R_i \cos(k(R_i + R_0)) \\ &\quad + R_0 A \sum X_i \sin(k(R_i + R_0)) - R_0 A^2 \sum \cos(k(R_i + R_0)) \sin(k(R_i + R_0)) \\ &\quad - R_0 B \sum Y_i \cos(k(R_i + R_0)) + R_0 B^2 \sum \sin(k(R_i + R_0)) \cos(k(R_i + R_0)) \\ &\quad + R_0 BC \sum \cos(k(R_i + R_0)) \\ &= A \sum X_i R_i \sin(k(R_i + R_0)) - A^2 \sum R_i \cos(k(R_i + R_0)) \sin(k(R_i + R_0)) \\ &\quad - B \sum Y_i R_i \cos(k(R_i + R_0)) + B^2 \sum R_i \sin(k(R_i + R_0)) \cos(k(R_i + R_0)) \\ &\quad + BC \sum R_i \cos(k(R_i + R_0)) \\ \frac{1}{2} \frac{\partial SEE^2}{\partial R_0} &= kA \sum X_i \sin(k(R_i + R_0)) - kA^2 \sum \cos(k(R_i + R_0)) \sin(k(R_i + R_0)) \\ &\quad - kB \sum Y_i \cos(k(R_i + R_0)) + kB^2 \sum \sin(k(R_i + R_0)) \cos(k(R_i + R_0)) \\ &\quad + kBC \sum \cos(k(R_i + R_0)) \end{aligned} \tag{2.16}$$

$$\begin{aligned}
A &= \frac{\sum X_i \cos(k(R_i + R_0))}{\sum \cos^2(k(R_i + R_0))} \\
B &= \frac{N \sum Y_i \sin(k(R_i + R_0)) - \sum Y_i \sum \sin(k(R_i + R_0))}{N \sum \sin^2(k(R_i + R_0)) - \sum \sin(k(R_i + R_0)) \sum \sin(k(R_i + R_0))} \\
C &= \frac{\sum Y_i \sum \sin^2(k(R_i + R_0)) - \sum Y_i \sin(k(R_i + R_0)) \sum \sin(k(R_i + R_0))}{N \sum \sin^2(k(R_i + R_0)) - \sum \sin(k(R_i + R_0)) \sum \sin(k(R_i + R_0))}
\end{aligned} \tag{2.17}$$

$$\begin{aligned}
i &= \arcsin \frac{-B}{A} \\
\phi_0 &= R_0 \cdot k \\
\theta &= \arctan \frac{A \cos i}{C} \\
\Omega_{\text{ref}} &= \left(\frac{A}{r_{\text{ref}} \sin \theta} \right)^{\frac{4}{3}} \frac{1}{\cos i} \\
T &= k \frac{\cos^{\frac{1}{4}} i}{\Omega_{\text{ref}}^{\frac{3}{4}}}
\end{aligned} \tag{2.18}$$

[(YN) The external part of the streamer where the infall term cannot be safely neglected should be truncated. The procedure should be repeated until convergence is obtained.]

2.5.3 Position angle

[(YN) This step can be neglected if the disk PA on the sky is known. Otherwise, the previous step needs to be repeated until the optimal is found.]

2.5.4 Fitting with partially known parameters

[(YN) If the inclination is known, fitting could be simplified. $B = -A \sin i$. Re-write the equations for the regression model: only 4 equations to be solved.]

Chapter 3

Results



Chapter 4

Discussion



Chapter 5

Conclusion



Appendix A

Observation

Table A.1: Observations. The inclination angle with $i=0$ corresponds to a face-on disk.

Target	RA (J2000) (h m s)	Decl (J2000) ($^{\circ}$ ' ")	Stellar mass ($M_{\odot}$)	Systemic velocity (km/s)	Disk inclination	Disk position angle	Reference
Per-emb-2	03 : 32 : 17.9320	30.49.47.7050	3.2				1
Per-emb-50	03 : 29 : 07.7680	31.21.57.1280	2.58		67°	170°	2
HLTau	04 : 31 : 38.4280	18.13.57.1900	2.1	7.14	47°	$^{\circ}$	3
S CrA	19 : 01 : 08.6000	-36.57.20.0000	2	5.86	65°	$^{\circ}$	4
SVS 13A	03 : 29 : 03.7400	31.16.03.8000			$^{\circ}$	$^{\circ}$	5

References. ⁽¹⁾Pineda et al. 2020 ⁽²⁾Valdivia-Mena et al. 2022 ⁽³⁾Yen et al. 2019
⁽⁴⁾Gahm et al. 2018 ⁽⁵⁾Hsieh et al. 2023

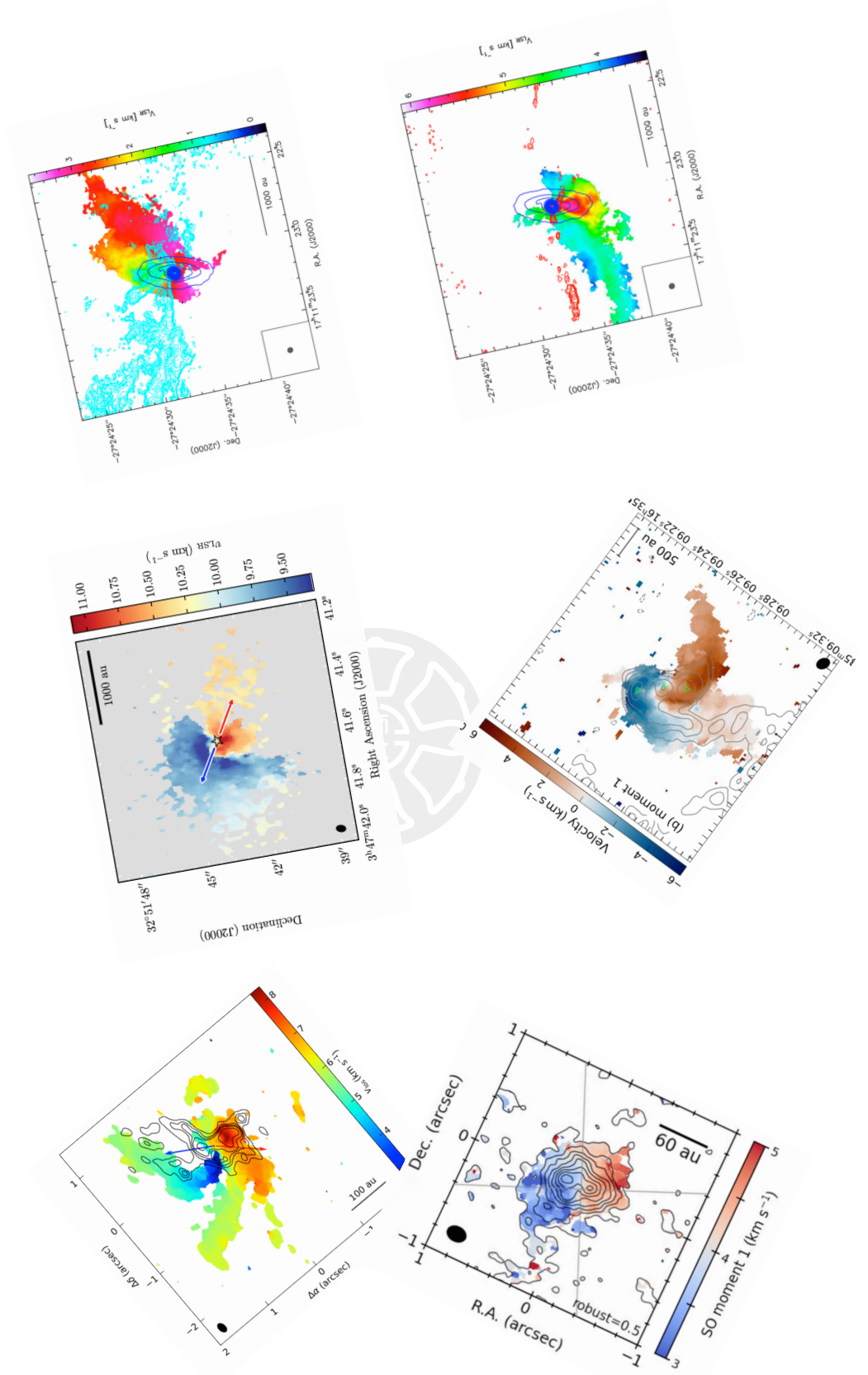


Figure A.1: Previous observations

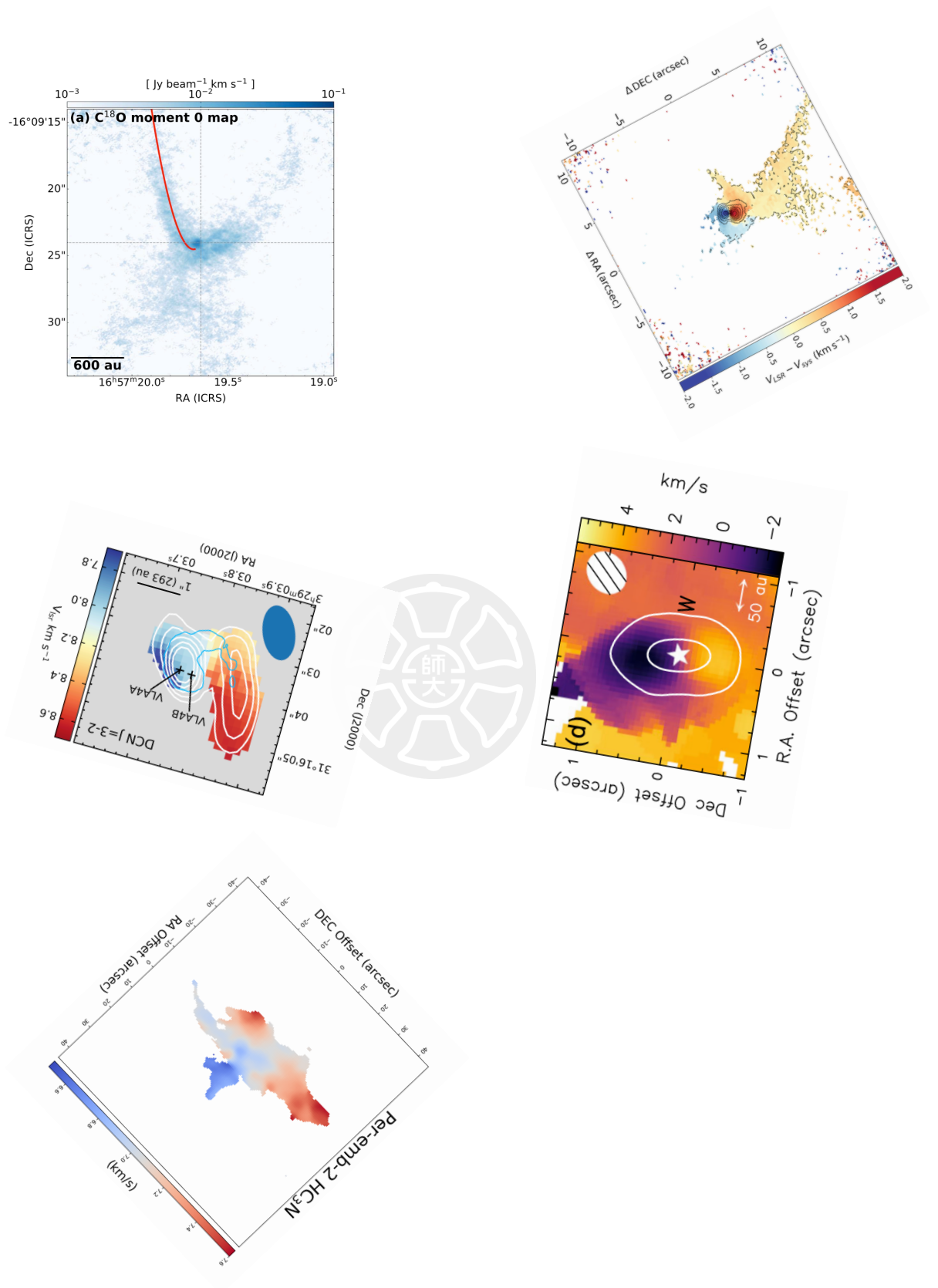


Figure A.2: Previous observations

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